FairFlow

A Food Distribution & Optimization App

PRESENTED AND MADE BY

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Introduction & Project Overview

The Idea~

FairFlow is a data driven demand forecasting and food distribution optimization app for local community pantries.

The Problem~

Food banks face an ongoing challenge in matching fluctuating food donations with community demand across different neighborhoods. Because demand varies weekly and depends on socioeconomic conditions, unemployment rates, and seasonal factors, many local pantries experience either food shortages or excess perishable inventory leading to spoilage. Currently, most food allocation decisions are manual and reactive, relying on prior experience rather than data analytics. This leads to inefficiencies, increased waste, and inequitable service coverage.

The Solution~

Predict neighborhood level food demand using historical distribution, census, and economic data. Next, recommend optimal allocation plans to minimize shortages and spoilage. Use the data to provide interactive dashboards for local pantry organizers to visualize demand hotspots and delivery routes.

Project Context

Scope of the issue ~

- 47M Americans face food insecurity, including 13M children (≈1 in 5 kids)
- USDA: 17M households affected in 2023 — sharp rise vs. prior years

Current challenges ~

- Lack of a strong central nationwide network
- Persistent inefficiencies in food collection, storage, and distribution

Prior Data-driven insights ~

- Second Harvest Heartland: used advanced mapping/logistics to improve targeting in Minnesota
- Feeding America's Map the Meal Gap: county/district disparities shaped by geography, economics, and demographics

FairFlow system ~

- Provide an intelligent, centralized platform for donors, food banks, volunteers, and recipients
- Use predictive analytics and data modeling to optimize operations and to forecast demand, align donations, reduce waste

Overall Mission ~

- Strengthen efficiency, transparency, and equity in food redistribution

System Requirements

Non-Functional Requirements

Performance ~
The system should load dashboards within a reasonable time. It should handle moderate to large datasets without crashing.

Security ~
User accounts must be protected through logins and role based access codes.
Data must be stored securely and transmitted over encrypted connections.

Usability ~
The interface should be simple enough for non-technical staff to use.
Visuals and dashboards should be easy to read and interpret.

Reliability ~
The system should be available with minimal downtime.
Data should be backed up regularly to avoid loss.

Scalability ~
The system should support more users, additional locations, or larger datasets over time.

Maintainability ~
The system should be modular so updates and fixes can be done easily.

Functional Requirements

Data Collection and Processing ~
The system must import and organize data from donation records, distribution logs, and partner agencies.

Demand Forecasting ~
The system must generate weekly or monthly predictions of food demand by region or partner site.

Allocation Recommendations ~
The system must recommend how available food should be distributed to different locations based on predicted need.

Dashboard and Visualizations ~
The system must provide charts, maps, and summaries that allow users to view demand, supply, and distribution activity.

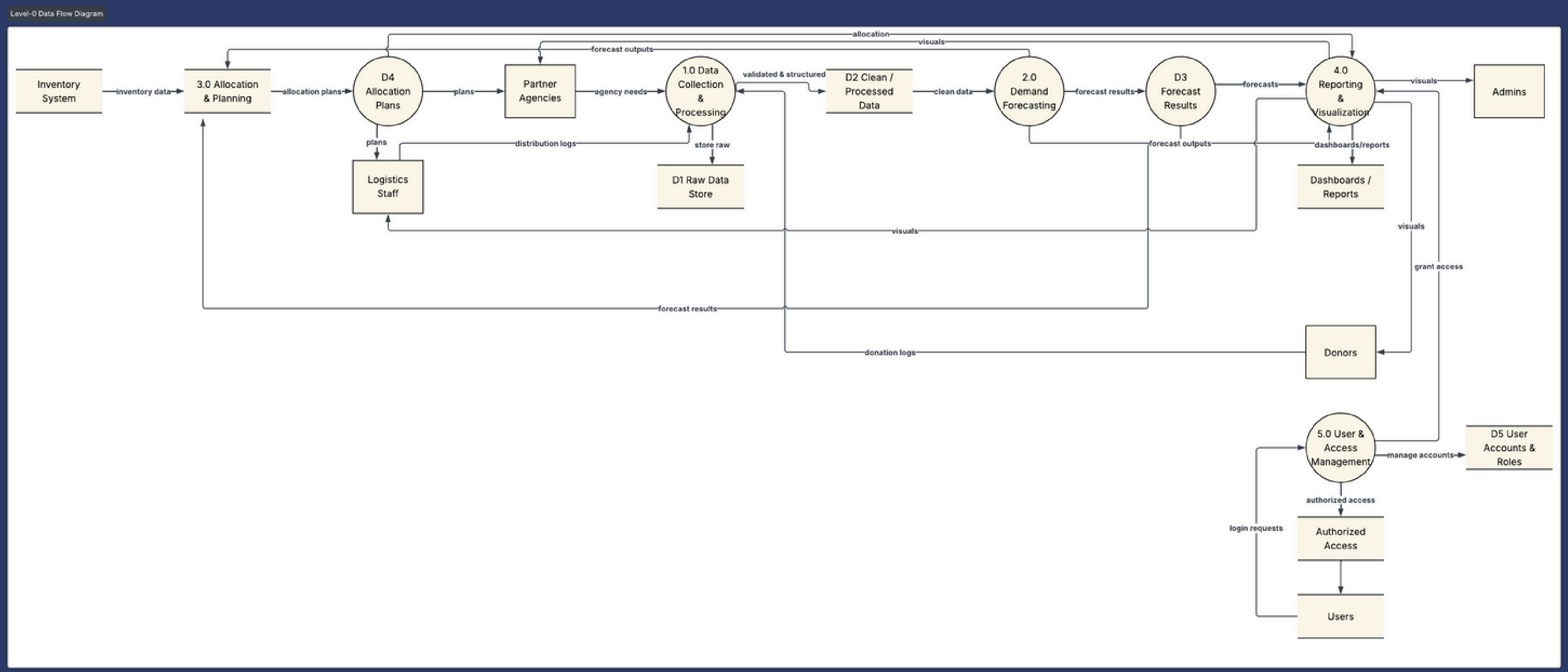
User Accounts and Permissions ~
The system must allow different user roles with appropriate access levels (enrolled individuals, admin, pantry staff , partner organization, etc).

Reporting Tools ~
The system must generate downloadable reports summarizing demand, allocations, and key metrics.

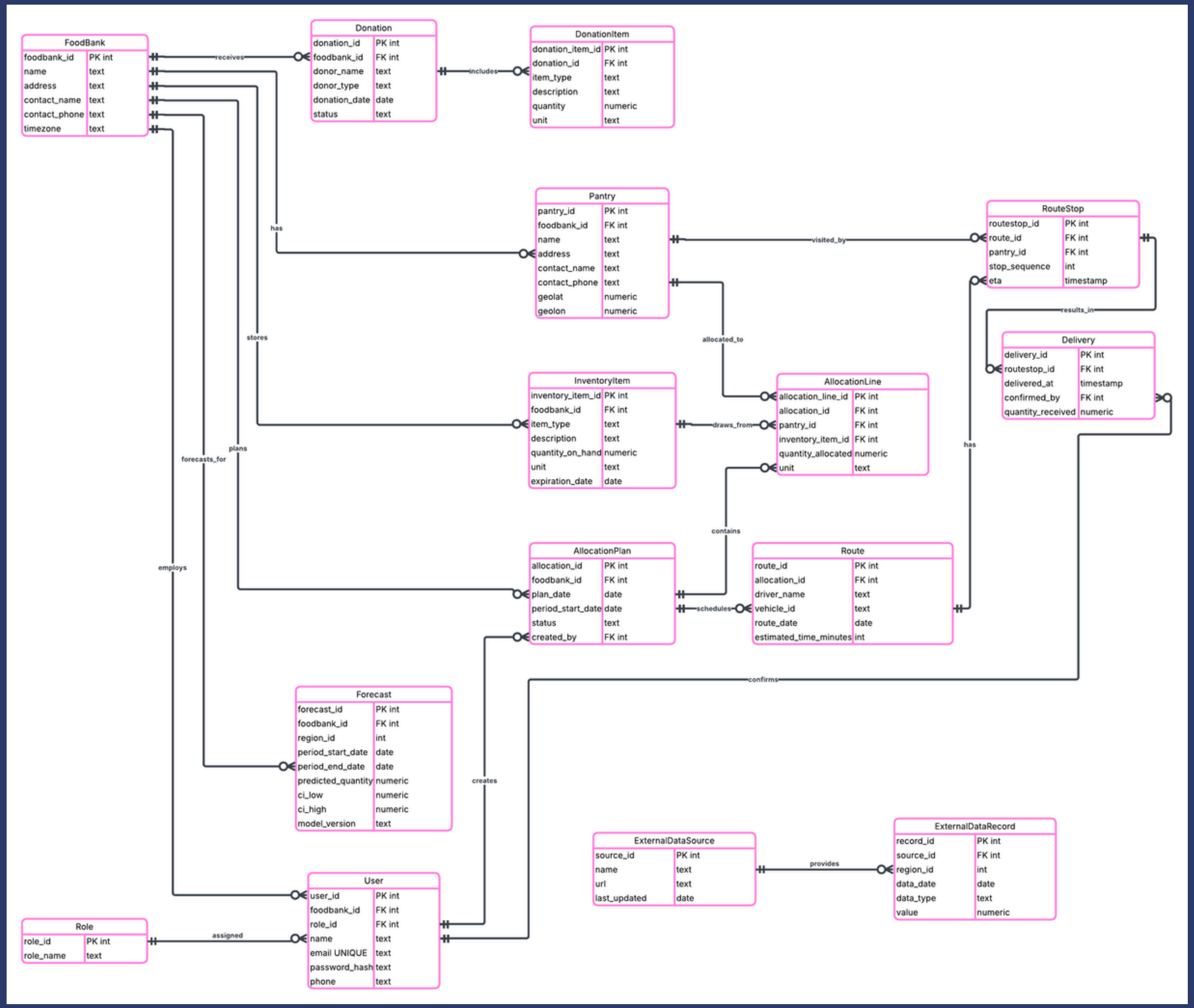
Notifications ~
The system must alert users to shortages, upcoming distribution needs, or data issues.

System Design & Data Modeling

Level-0 Data Flow Diagram

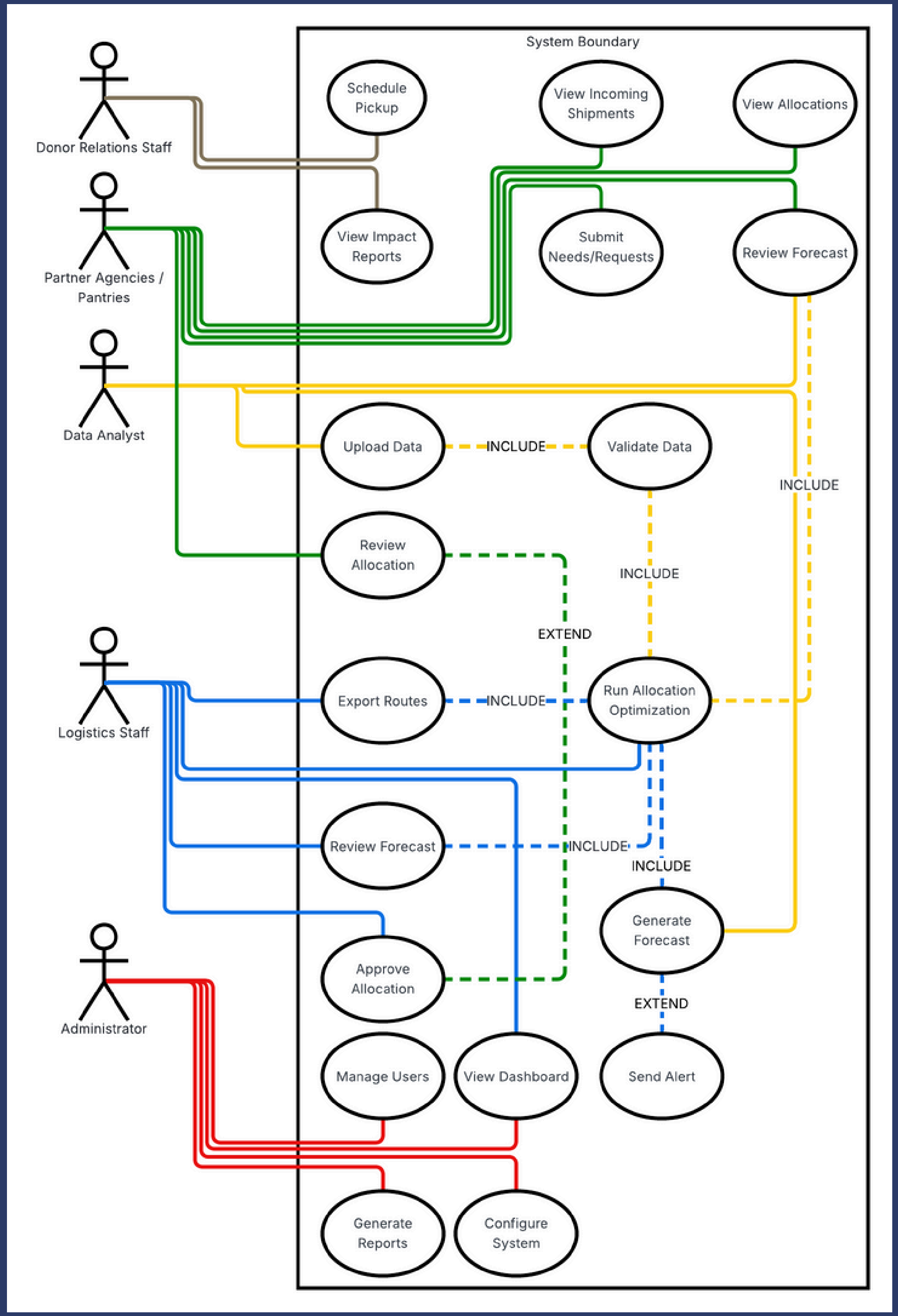


ERD using Crowsfoot Notation



System Design & Data Modeling 2

Use Case Diagram



Processes

Collect and Validate Data ~

Gathers pantry requests, inventory updates, and donor information, then checks for completeness and accuracy.

Generate Forecasts ~

Uses validated data and historical trends to estimate future food demand.

Optimize Allocation ~

Determines how food should be distributed based on forecasts, inventory, and constraints.

Manage Routing and Distribution ~

Converts allocation plans into delivery routes, sends assignments to drivers, and records delivery confirmations.

Reporting and Dashboards ~

Compiles system data into summaries and visual reports for administrators, analysts, donors, and partner agencies.

System Architecture

FairFlow uses a cloud based three tier architecture consisting of a presentation layer, an application layer, and a data layer. This architecture was chosen because it is easy to scale, supports multiple users, and allows the system to handle forecasting, reporting, and data processing without slowing down. It also separates responsibilities, making the system easier to maintain and update over time.

System Architecture

Proposed System Architecture ~

Presentation Layer (Client Side) ~

- Web browser and app interface used by administrators, logistics staff, data analysts, donors, and partner pantries.
- Displays dashboards, maps, reports, inventory, deliveries and allocation recommendations.

Application Layer (Server Side) ~

- Handles business logic, demand forecasting, and allocation calculations.
- Provides APIs for the interface to request data and submit updates.

Data Layer (Storage) ~

- Central database that stores inventory, pantry requests, forecasts, deliveries and reports.
- File storage for uploaded data, historical reports, and model outputs.

Hardware & Software Specifications

Software ~

- Frontend: App (JavaScript) Web application (HTML, CSS, JavaScript).
- Backend: Server side application (Python).
- Database: MySQL.
- Cloud Hosting: AWS or Azure

Hardware ~

- Cloud server (application layer):
- 2–4 vCPUs, 8–16 GB RAM.
- Database server:
- Managed cloud database, 200–300 GB storage.
- File storage:
- Cloud storage bucket for reports and data uploads.

Functional UI Prototype Created with Figma

[FairFlow Figma Prototype Link]

Conclusion

FairFlow offers a practical, scalable solution to help food banks and pantries distribute food more equitably and efficiently. By combining demand forecasting, optimized allocation, and transparent dashboards, it reduces waste, improves coverage in high-need areas, and strengthens donor trust, thus laying the foundation for a more fair, data driven food distribution system.

FairFlow

Thank You!